Amplitude-Phase Description of Stochastic Neural Oscillators: A complex-valued approach

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I. INTRODUCTION

II. STOCHASTIC WILSON-COWAN MODEL

We consider a stochastic version [1] of the seminal Wilson-Cowan model [2] for oscillation generation. The two-dimensional model with additive noise mimics a population of excitatory (E) and of inhibitory (I) cells. The mean synaptic coupling from E cells to themselves, and from E cells to I cells, are given respectively by W_{EE} and W_{IE}. Similarly, the mean synaptic coupling from I cells to themselves and from I cells to E cells are W_{II} and W_{EI}. The E cell population receives a constant external stimulus h_E, while the I cell population is driven by the constant external current h_I. Additive noise is assumed to arise mainly from random synaptic events. Defining the fraction of excitatory cells that are firing at a given time by E(t), and of inhibitory cells I(t), yields the stochastic WC model (SWC):

$$\frac{dE(t)}{dt} = -\alpha_E E(t) + \big(1 - E(t)\big)\beta_E f(s_E(t)) + \sigma_E \xi_E(t) \tag{1}$$

$$\frac{dI(t)}{dt} = -\alpha_I I(t) + \big(1 - I(t)\big)\beta_I f(s_I(t)) + \sigma_I \xi_I(t)\,. \tag{2}$$

The sigmoid function is $f(x) = (1+e^{-x})^{-1}$ and the total excitatory $s_E(t)$ and inhibitory $s_I(t)$ synaptic inputs to a neuron are given respectively by

$$s_E(t) = W_{EE}E(t) - W_{EI}I(t) + h_E \tag{3}$$

$$s_I(t) = W_{IE}E(t) - W_{II}I(t) + h_I\,. \tag{4}$$

where $\xi_{E,I}$ are two independent Gaussian white noises with the following properties:

$$< \xi_{E,I}(t) > = 0, \quad < \xi_E(t)\xi_I(t) >= 0$$
$$< \xi_E(t^{'})\xi_E(t) > =< \xi_I(t^{'})\xi_I(t) >= \delta(t - t^{'})\,.$$

The noise-free ($\sigma_E = \sigma_I = 0$) dynamics has a fixed point (E_0, I_0) (for the parameter values choose in this work). If the fixed point is a stable focus, deviations from the fixed point converge to zero. The noise slowly erases the memory of the initial conditions, and induces deviations from the fixed point that relax in a noisy oscillatory manner [3, 4]. Adding noise to this regime leads to noise-induced oscillations or quasi-cycles.

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III. AMPLITUDE-PHASE DECOMPOSITION FOR NETWORKS

Finally we extend our theory to interconnected networks of E-I units described by the SWC3 or SSL nonlinear rate models. One application of such amplitude-phase decomposition may be the study of the functional connectivity between brain areas [5–7]. We only consider the case where the network frequencies belong to the same band and therefore avoid the case of cross-frequency coupling where faster rhythms are coupled with slower ones [8].

A. Network of Stochastic Wilson-Cowan units

We consider a network of excitatorily coupled SWC3 units:

$$\frac{dE_k(t)}{dt} = -\alpha_E E_k + \big(1 - E_k\big)\beta_E f(s_{E_k}) + \sigma_{E_k}\xi_{E_k}(t) \tag{5}$$

$$\frac{dI_k(t)}{dt} = -\alpha_I I_k + \big(1 - I_k\big)\beta_I f(s_{I_k}) + \sigma_{I_k}\xi_{I_k}(t) \tag{6}$$

The total excitatory $s_{E_k}(t)$ and inhibitory $s_{I_k}(t)$ synaptic inputs to a neuron population are given by

$$s_{E_k}(t) = W^k_{EE}E_k - W^k_{EI}I_k + h^k_E + \sum_{l=1}^{N} S^{kl}_{EE}E_l(t - \tau_{kl}) \tag{7}$$

$$s_{I_k}(t) = W^k_{IE}E_k - W^k_{II}I_k + h^k_I + \sum_{l=1}^{N} S^{kl}_{IE}E_l(t - \tau_{kl})\,. \tag{8}$$

The coefficient S^{kl}_{EE} denotes the long-range excitatory weight from excitatory population k to excitatory population l, while S^{kl}_{IE} accounts for the weight from excitatory population k to inhibitory population l. As for the case of the single E-I units, we look at the fluctuations of each unit as follows:

$$E_k(t) = E^k_0 + V_{E_k}(t) \quad and \quad I_k(t) = I^k_0 + V_{I_k}(t)\,. \tag{9}$$

Each basic E-I units behaves as above, but for simplicity, we keep only the linear coupling terms, and neglect all nonlinear terms arising from couplings between units. Therefore, the only difference is the fact that the coefficients of the single unit analysis now have the index k

which characterizes the heterogeneity of each network. But the form of the coefficients of the deviations dynamics are similar. Specifically, the fixed points (E_0^k, I_0^k) now depend on the coupling terms. The fluctuation dynamics are given by the $2N$-dimensional system:

$$\frac{dV_{E_k}}{dt} = A_{EE}^k V_{E_k} + A_{EI}^k V_{I_k} + \sum_{l=1}^{N} C_{EE}^{kl} V_{E_l}(t-\tau_{kl}) + \sigma_{E_k}\xi_{E_k}(t) \tag{10}$$

$$\frac{dV_{I_k}}{dt} = A_{IE}^k V_{E_k} + A_{II}^k V_{I_k} + \sum_{l=1}^{N} C_{IE}^{kl} V_{E_l}(t-\tau_{kl}) + \sigma_{I_k}\xi_{I_k}(t)\,. \tag{11}$$

with the following parameter definitions:

$$\begin{aligned}
A_{EE}^k &= -\big[\alpha_E + \beta_E f(s_{E_0^k}) - (1-E_0^k)\beta_E f^{'}(s_{E_0^k}) W_{EE}^k\big] \\
A_{II}^k &= -\big[\alpha_I + \beta_I f(s_{I_0^k}) + (1-I_0^k)\beta_I f^{'}(s_{I_0^k}) W_{II}^k\big] \\
A_{EI}^k &= -(1-E_0^k)\beta_E f^{'}(s_{E_0^k}) W_{EI}^k \\
A_{IE}^k &= (1-I_0^k)\beta_I f^{'}(s_{I_0^k}) W_{IE}^k\,.
\end{aligned}$$

The effective coupling coefficients C_{EE}^{kl} and C_{EE}^{kl} are:

$$\begin{aligned}
C_{EE}^{kl} &= (1-E_0^k)\beta_E f^{'}(s_{E_0^k}) S_{EE}^{kl} \\
C_{IE}^{kl} &= (1-I_0^k)\beta_I f^{'}(s_{I_0^k}) S_{IE}^{kl}
\end{aligned} \tag{12}$$

with the following expressions for the inputs to each population of the k-th unit:

$$\begin{aligned}
s_{E_0^k} &= W_{EE}^k E_0^k - W_{EI}^k I_0^k + h_E^k + \sum_{l=1}^{N} S_{EE}^{kl} E_0^l \\
s_{I_0^k} &= W_{IE}^k E_0^k - W_{II}^k I_0^k + h_I^k + \sum_{l=1}^{N} S_{IE}^{kl} E_0^l\,.
\end{aligned}$$

Here $(E_0^k, I_0^k), k = 1...N$ are the fixed points of the stationary noise-free Eqs. (5-6). The coefficients of Eqs. (10-11) have the same form as in the single unit case; however the expressions for (s_{E_0}, s_{I_0}) are now replaced by $(s_{E_0^k}, s_{I_0^k}, k = 1...N)$. Similarly, (E_0, I_0) is replaced by $(E_0^k, I_0^k, k = 1...N)$.

As in the case of the single unit, linear stability analysis suggests a fixed amplitude ratio and phase difference between the E and I components of each unit. However, the determination of these quantities is different from the single unit case. To find the frequency, amplitude ratio and phase difference, these steps are followed:

- From the linear stability analysis, compute the eigenvalues of the corresponding $2N$-dimensional linear system associated with Eqs. (10-11). The number of eigenvalues is infinite when the propagation is consider and equal to $2\times N$ otherwise. The imaginary part ω_0 of the eigenvalue with the largest real part is considered as the mean frequency of the system.

- Consider the eigenvalue with the largest real part $(\lambda = \nu_{max} + j\omega_0)$ and solve the following $N \times N$ system of equations for the amplitude ratios derived from the linear stability analysis:

$$A_{EI}^k \tilde{\alpha}_{kk} + \sum_{l=1}^{N} C_{EE}^{kl} \exp(-\lambda\tau_{kl})\tilde{\alpha}_{kl} = \lambda - A_{EE}^k$$

$$\sum_{l=1}^{N} C_{IE}^{kl} \exp(-\lambda\tau_{kl})\tilde{\alpha}_{kl} + (A_{II}^k - \lambda)\tilde{\alpha}_{kk} = -A_{IE}^k$$

$$\tilde{\alpha}_{kl} = \big(\tilde{\alpha}_{lk}\big)^{-1}, \quad k, \neq l = 1, ..., N$$

  where

$$\tilde{\alpha}_{kk} = \frac{\tilde{B}_{I_k}}{\tilde{B}_{E_k}}, \quad \tilde{\alpha}_{kl} = \frac{\tilde{B}_{E_l}}{\tilde{B}_{E_k}}, \quad k, \neq l = 1, ..., N\,. \tag{13}$$

- The amplitude ratio and phase difference between inhibitory and excitatory fluctuations are therefore given by:

$$\alpha_k = \big|\tilde{\alpha}_{kk}\big| \quad ; \delta_k = arg\big(\tilde{\alpha}_{kk}\big),\ k = 1, ..., N$$

  where $||$ and arg are respectively the modulus and the argument of a complex number.

As in the case of single unit, we can look for solutions of excitatory and inhibitory fluctuation dynamics for each population in each unit in the form:

$$\begin{aligned}
V_{E_k}(t) &= r_k(t)\cos\big(\omega_0 t + \phi_k(t)\big) \\
V_{I_k}(t) &= \alpha_k r_k(t)\cos\big(\omega_0 t + \phi_k(t) + \delta_k\big)\,.
\end{aligned} \tag{14}$$

We have dropped the subscript E for the variables r and ϕ for the sake of readability. We then plugged these expressions in the system of $2N$-dimensional Eqs. (10-11) and obtained the final system of $2N$-dimensional equations for the amplitudes r_k and phases ϕ_k:

$$\frac{dr_k}{dt} = F_1^k\big(r_k, \phi_k\big) + G_1^k\big(r_k, \phi_k, \xi_{E_k}, \xi_{I_k}\big) \tag{15}$$

$$\frac{d\phi_k}{dt} = F_2^k\big(r_k, \phi_k\big) + G_2^k\big(r_k, \phi_k, \xi_{E_k}, \xi_{I_k}\big) \tag{16}$$

with the following functions:

$$
\begin{aligned}
F_1^k(r_k,\phi_k) &= \frac{1}{\alpha_k \sin(\delta_k)}\Big[\alpha_k f_1^k(r_k,\phi_k)\sin(\omega_0 t+\phi_k+\delta_k) - f_2^k(r_k,\phi_k)\sin(\omega_0 t+\phi_k)\Big] \\
F_2^k(r_k,\phi_k) &= \frac{1}{\alpha_k r_k \sin(\delta_k)}\Big[\alpha_k f_1^k(r_k,\phi_k)\cos(\omega_0 t+\phi_k+\delta_k) - f_2^k(r_k,\phi_k)\cos(\omega_0 t+\phi_k)\Big] \\
G_1^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k}) &= \frac{1}{\alpha_k \sin(\delta_k)}\Big[\alpha_k g_1^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k})\sin(\omega_0 t+\phi_k+\delta_k) - g_2^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k})\sin(\omega_0 t+\phi_k)\Big] \\
G_2^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k}) &= \frac{1}{\alpha_k r_k \sin(\delta_k)}\Big[\alpha_k g_1^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k})\cos(\omega_0 t+\phi_k+\delta_k) - g_2^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k})\cos(\omega_0 t+\phi_k)\Big] \\
f_1^k(r_k,\phi_k) &= \omega_0 r_k \sin(\omega_0 t+\phi_k) + A_{EE}^k r_k \cos(\omega_0 t+\phi_k) + A_{EI}^k \alpha_k r_k \cos(\omega_0 t+\phi_k+\delta_k) \\
&\quad + \sum_{l=1}^{N} C_{EE}^{kl} r_l(t-\tau_{kl})\cos[\omega_0 t - \omega_0\tau_{kl} + \phi_l(t-\tau_{kl})] \\
f_2^k(r_k,\phi_k) &= \alpha_k\omega_0 r_k \sin(\omega_0 t+\phi_k+\delta_k) + A_{IE}^k r_k \cos(\omega_0 t+\phi_k) + A_{II}^k \alpha_k r_k \cos(\omega_0 t+\phi_k+\delta_k) \\
&\quad + \sum_{l=1}^{N} C_{IE}^{kl} r_l(t-\tau_{kl})\cos[\omega_0 t - \omega_0\tau_{kl} + \phi_l(t-\tau_{kl})] \\
g_1^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k}) &= \sigma_{E_k}\xi_{E_k}(t); \quad g_2^k(r_k,\phi_k,\xi_{E_k},\xi_{I_k}) = \sigma_{I_k}\xi_{I_k}(t).
\end{aligned}
$$

1. *Stochastic Averaging Method (SAM)*

To simplify the amplitude-phase dynamics in Eqs. 15 and 16, we make use of the Stochastic Averaging Method (SAM) [9], which stipulates that the system can be approximated by a two-dimensional Markov process given by:

$$
d\begin{pmatrix} Z_E(t) \\ \phi_E(t) \end{pmatrix} = \begin{pmatrix} m_1(t) \\ m_2(t) \end{pmatrix} dt + \begin{pmatrix} h_{11}(t) & h_{12}(t) \\ h_{21}(t) & h_{22}(t) \end{pmatrix} \begin{pmatrix} dW_1(t) \\ dW_2(t) \end{pmatrix}
$$

$$
m = T^{av}\left(E\{F\} + \int_{-\infty}^{0} E\left\{ \left(\frac{\partial G}{\partial X}\right)_t (G)_{t+\tau} \right\} d\tau \right)
$$

$$
h\,h' = T^{av}\left(\int_{-\infty}^{\infty} E\{(G)_t (G')_{t+\tau}\} d\tau \right).
$$

Here $m = \begin{pmatrix} m_1(t) \\ m_2(t) \end{pmatrix}$, $F = \begin{pmatrix} F_1(t) \\ F_2(t) \end{pmatrix}$, $G = \begin{pmatrix} G_1(t) \\ G_2(t) \end{pmatrix}$, h is the matrix made up of the elements h_{ij}, $(h\,h')$ is the product of h and its transpose, and

$$
\left(\frac{\partial G}{\partial X}\right)_t \equiv \begin{pmatrix} \frac{\partial G_1}{\partial r_E} & \frac{\partial G_1}{\partial \phi_E} \\ \frac{\partial G_2}{\partial r_E} & \frac{\partial G_2}{\partial \phi_E} \end{pmatrix}
$$

is a 2x2 Jacobian matrix. Moreover, $E.$ denotes the expectation operator (i.e. average over realizations), and T^{av} is the time averaging operator defined by

$$
T^{av}(.) = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} (.)\,dt\,.
$$

The subscript t and $t+\tau$ mean that the elements of the corresponding vectors and matrix are evaluated at these values. The SAM further assumes that the 2-dimensional functions $F(r_E,\phi_E)$ and $G(r_E,\phi_E,\xi_E,\xi_I)$ are respectively of order $\mathrm{O}(\epsilon)$ and $\mathrm{O}(\epsilon^2)$, where ϵ is a small parameter. The averaging in the SAM procedure assumes that the functions $r_E(t)$ and $\phi_E(t)$ are constant during the period $T_0 = 2\pi/\omega_0$ of the underlying deterministic oscillation (ω_0 also varies with system parameters).

Applying the SAM method to the $2N$-dimensional system (Eqs. 15-16) leads to the following amplitude-phase equations for the networks of SWC units:

with the definitions

We have thus derived an amplitude-phase decomposition of connected SWC E-I units through the SAM by extending the technique used for an isolated E-I network. The result is a set of $2\times N$ stochastic differential equations for the amplitudes and phases of each unit. The connectivity between the amplitude and phase are made through sinusoidal coupling functions. Unfortunately, the amplitude equation is not uncoupled from that for the phase as was the case for a single unit. A Fokker-Planck analysis is therefore difficult to conduct.

To recover the common fast phase variable usually used and associated to the Hilbert transform, it suffices to consider the change of variable $\theta_k(t) = \omega_0 t + \phi_k(t)$. This leads to the following amplitude-phase dynamics

$$dr_k(t) = \Big\{ -\lambda_k r_k(t) + \frac{D_k}{2r_k(t)} + \sum_{l=1}^{N} \Big[K_{EE}^{kl} \sin\big(\theta_l(t-\tau_{kl}) - \theta_k(t) - \delta_k\big) - K_{IE}^{kl} \sin\big(\theta_l(t-\tau_{kl}) - \theta_k(t)\big) \Big] r_l(t-\tau_{kl}) \Big\} dt \tag{17}$$

$$+ \sqrt{D_k}\, dW_1^k(t)$$

$$d\theta_k(t) = \Big\{ \omega_k - \sum_{l=1}^{N} \Big[K_{EE}^{kl} \cos\big(\theta_l(t-\tau_{kl}) - \theta_k(t) - \delta_k\big) - K_{IE}^{kl} \cos\big(\theta_l(t-\tau_{kl}) - \theta_k(t)\big) \Big] \frac{r_l(t-\tau_{kl})}{r_k(t)} \Big\} dt + \frac{\sqrt{D_k}}{r_k(t)} dW_2^k(t)\,. \tag{18}$$

The new parameter ω_k is related to Ω_k and depends on the intrinsic parameter of each network, as following

$$\lambda_k = -\frac{A_{EE}^k + A_{II}^k}{2}$$
$$\omega_k = \frac{\alpha_k \cos(\delta_k)(A_{EE}^k - A_{II}^k) + \alpha_k^2 A_{EI}^k - A_{IE}^k}{2\alpha_k \sin(\delta_k)}$$
$$D_k = \frac{(\alpha_k \sigma_{E_k})^2 + (\sigma_{I_k})^2}{2\alpha_k^2 \sin^2(\delta_k)}$$
$$K_{EE}^{kl} = \frac{\alpha_k C_{EE}^{kl}}{2\alpha_k \sin(-\delta_k)}$$
$$K_{IE}^{kl} = \frac{C_{IE}^{kl}}{2\alpha_k \sin(-\delta_k)}$$

It is related to the deterministic frequency of the oscillator k and coincides with the imaginary part of the complex conjugate eigenvalues ω_0 for a single oscillator. However, we found that the complete deterministic frequency is generally amplitude-dependent as shown by the two first terms in right side of the equality of Eq. 18. The news dynamics Eqs.(17-18) represent a general model for the interaction of oscillators in the slow $r_k(t)$ and fast $\theta_k(t)$ variables framework.

B. Complex coordinate representation

A more appropriate way to represent the transient dynamics of the E and I oscillations is through their analytical representations. In this framework, the signals is described in the complex plane with the real part being the actual signals and the imaginary parts their Hilbert transforms. We defined the analytical signal of the E variable as follows:

$$x_{E_k}(t) = r_k \exp(j\theta_k) \tag{19}$$

Taking the derivative leads to the following

$$dx_{E_k} = dr_k \exp(j\theta_k) + r_k d(\exp(j\theta_k)) \tag{20}$$

We now replace the dr_k by its expression from Eq.17 and use the change of variable formula from Ito calculus to derive an expression of $d(\exp(j\theta_k))$ from Eq.18, and obtain the following first-order complex stochastic differential equation

$$\dot{x}_{E_k}(t) = (-\lambda_k + j\omega_k) x_{E_k}(t) + \sum_{l=1}^{N} e^{-j\phi_{kl}} a_{kl} x_{E_l}(t-\tau_{kl}) + \sqrt{D_k}\xi_k(t) \tag{21}$$

where the real coefficients a_{kl} and ϕ_{kl} describe the complex coupling among the oscillators through the following relation

$$e^{-j\phi_{kl}} a_{kl} = e^{-j\frac{\pi}{2}} \big(e^{-j\delta_k} K_{EE}^{kl} - K_{IE}^{kl} \big) \tag{22}$$

1. *Special cases*

- If the long-range coupling to excitatory neurons is null ($K_{EE}^{kl} = 0$) then we have the following:

$$a_{kl} = -K_{IE}^{kl}; \qquad \phi_{kl} = \phi = -\frac{\pi}{2}$$

- If the long-range coupling targets only excitatory neurons ($K_{IE}^{kl} = 0$), we have the following

$$a_{kl} = K_{EE}^{kl}, \qquad \phi_{kl} = \phi_k = \frac{\pi}{2} - |\delta_k|$$

2. *Homogeneous case without propagation delays*

If we consider a homogeneous case and consider that information travels instantaneous from one oscillator to the other, the complex-valued equation Eq.21 reduces to

$$\dot{x}_{E_k}(t) = (-\lambda_0 + j\omega_0) x_{E_k}(t) + \sum_{l=1}^{N} e^{-j\phi_{kl}} a_{kl} x_{E_l}(t) + \sqrt{D_k}\xi_k(t) \tag{23}$$

In matrix representation the equation becomes

$$\dot{\vec{x}}_E(t) = \big(diag(-\lambda_0 + j\omega_0) + e^{-j\phi} a \big) \vec{x}_E(t) + \sqrt{D}\xi(t) \tag{24}$$

where, ϕ and a are matrices with coefficients ϕ_{kl} and a_{kl} respectively.

We are interested in only the stable solutions of Eq.**??**. When the matrix K admits positive eigenvalues with λ_{max} the real part of the largest among them, the stability condition is expressed as follows

$$\lambda_0 \in \left[\lambda_{max}, +\infty\right[\tag{25}$$

IV. DISCUSSION

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